RECONFIGURATION OF HOMOMORPHISMS TO TRIANGLE-FREE REFLEXIVE GRAPHS

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Abstract. For a graph H, the H-recolouring problem $\mathrm{Recol}(H)$ asks for two given homomorphisms from a given G to H, if one can get between them by a sequence of homomorphisms of G to H in which consecutive homomorphisms differ on only one vertex. We show that, if G and H are reflexive and H is triangle-free, then this problem can be solved in polynomial time. This also shows, at the same time, that the closely related H-reconfiguration problem $\mathrm{Recon}(H)$ of deciding whether two given homomorphisms from a given graph G to H are in the same component of the Hom-graph $\mathrm{Hom}(G, H)$, can be solved in polynomial time for triangle-free reflexive graphs H.

1. Introduction

Reconfiguration, in various settings, is the common notion of moving between solutions of a combinatorial problem via small changes; see van den Heuvel [?] and Nishimura [?] for detailed surveys. For example, a reconfiguration, or *recolouring*, between two graph colourings is a sequence of colourings in which consecutive elements differ on one vertex. The *k-recolouring problem* is to decide, for two given k-colourings of G, whether or not there is a recolouring between them. Cereceda, van den Heuvel and Johnson [?] showed that 3-recolouring is polynomial time solvable, while Bonsma and Cereceda [?] showed that k-recolouring is PSPACE-complete for $k \geq 4$. One should compare this to the k-colouring problem which is well known to be polynomial time solvable for $k = 2$ but NP-complete for $k \geq 3$.

The k-colouring problem generalizes to the H-colouring problem $\mathrm{Col}(H)$ for a graph H and to the constraint satisfaction problem $\mathrm{CSP}(\mathcal{H})$ for a relational structure $\mathcal{H}$. The computational complexity of $\mathrm{Col}(H)$, and its myriad of generalizations and variations, have been well studied over the last 50 years, culminating in the recent CSP dichotomy theorem of Bulatov [?] and Zhuk [?], which says that $\mathrm{CSP}(\mathcal{H})$ is either polynomial time solvable or NP-complete for every relational structure $\mathcal{H}$.

Analogously, the k-recolouring problem also generalizes to reconfiguration problems for H-colourings and CSP, both of which are well studied; see, e.g., [? ? ? ? ?] and [? ? ? ? ? ? ? ?], respectively. In particular, we mention that Gopalan et al. [?] proved a trichotomy theorem for the reconfiguration variation

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of CSP($\mathcal{H}$) for structures $\mathcal{H}$ with two vertices. In this paper, we focus on the complexity of H-colouring reconfiguration in the setting of graphs which are *reflexive*, in the sense that they have a loop on every vertex; graphs with no loops at all are called *irreflexive*. While such problems are also interesting for digraphs [**?**], in this paper we restrict our attention mainly to symmetric (i.e. undirected) graphs.

Recall that a *homomorphism* $\phi : G \to H$ from a graph G to a graph H is an edge-preserving vertex map. It is also known as an *H-colouring* of G due to the fact that a K_k-colouring of G, for the irreflexive clique K_k, is equivalent to a k-colouring. A *recolouring* between two H-colourings of G is a sequence of H-colourings in which consecutive elements differ on one vertex. An instance (G, ϕ, ψ) of the *H-recolouring problem* Recol(H) consists of a graph G and two H-colourings of G; the task is to decided whether or not there is a recolouring between them.

The *Hom-graph* Hom(G, H) for two graphs G and H has the set of all homomorphisms from G to H as its vertex set, and two homomorphisms ϕ and ψ are adjacent if, for all edges uv in G, $\phi(u)\psi(v)$ is an edge in H. It has been observed by several authors (e.g. [**?** **?**]) that, for undirected graphs, the H-recolouring problem is essentially the same as asking whether there is a path between a given pair of homomorphisms in Hom(G, H); we call this the *H-reconfiguration problem* and denote it by Recon(H). In one direction, any path from ϕ to ψ in Hom(G, H) can be converted into a path in which any two consecutive H-colourings differ on one vertex by simply changing the colours "one at a time." In the other direction, if (G, ϕ, ψ) is a YES instance for Recol(H), then the sequence of colourings is a path from ϕ to ψ in Hom(G, H) unless there exists an isolated vertex v of G with a loop which is mapped to different components of H by ϕ and ψ.[1] While we will state our results in terms of Recol(H), it is often convenient to speak in terms of paths in the graph Hom(G, H) for the sake of exposition.

The diameter of Hom(G, H) can be superpolynomial in $|V(G)|$ (see, e.g., [**?**]) and so, a priori, a YES instance of Recol(H) may not have a certificate of polynomial size. Thus, the natural complexity class for Recol(H) is PSPACE, rather than NP.

Some of the known results for irreflexive graphs are summarized as follows. The case that H is a clique follows from the results of [**?** **?**] for k-recolouring discussed earlier. In [**?**], the dichotomy for cliques was generalized to 'circular cliques' $G_{p/q}$; specifically, Recol($G_{p/q}$) is polynomial time solvable if $p/q < 4$ and PSPACE-complete if $p/q \geq 4$. The polynomial time algorithm for 3-recolouring in [**?**] had a topological flavor which Wrochna distilled and extended in [**?**] to prove the very general result that Recol(H) is polynomial time solvable for any graph H not containing a 4-cycle. In [**?**], we proved that, if H is a $K_{2,3}$-free quadrangulation of the sphere different from the 4-cycle, then Recol(H) is PSPACE-complete.

In the current paper, we use several ideas of these earlier papers, and adapt Wrochna's language and many of his ideas, to prove an analogue of his result for reflexive graphs G and H. Reflexive graphs are actually actually a smoother setting for some of these ideas; e.g. the fact that two consecutive vertices in a path in G can map to the same vertex of H eliminates a parity obstruction which turns up in [**?**]. We prove the following.

⟨thm:main⟩ **Theorem 1.1.** *If H is a reflexive triangle-free graph, then* Recol(H) *can be solved in polynomial time for reflexive instances.*

[1] In contrast, it is interesting to remark that the problems Recol(H) and Recon(H) do not have the same complexity in general in the case of digraphs [**?**].

In contrast, we recently proved in [**?**] that Recol(H) is PSPACE-complete when H is any reflexive triangulation of the sphere, other than a triangle itself. We refer to this paper for a deeper suggestion of the connection of topology to the complexity of the reconfiguration problem. We remark that the argument of Wrochna [**?**] directly generalizes to get that Recol(H) is solvable in polynomial time for general instances if H is a graph which may have loops on some vertices which does not contain a 4-cycle, a triangle with a loop on one vertex, or two adjacent vertices with loops; see the footenote on p. 330 of [**?**].

Before we move on to the proof of the main theorem, we observe that it implies the following result for a less general class of graphs H, but more general instances. Recall that the *girth* of a graph is the length of its shortest cycle.

Corollary 1.2. *If H is a reflexive symmetric graph with girth at least* 5*, then* Recol(H) *can be solved in polynomial time for all instances.*

Proof. Let (G, ϕ, ψ) be an instance of Recol(H). If G has isolated vertices, then we can simply change the colour of every such vertex v from $\phi(v)$ to $\psi(v)$ at the beginning without any issues. So, we assume that G has no isolated vertices.

Let G' be the graph obtained from G by adding all possible loops. We claim that the instance (G', ϕ, ψ) of Recol(H) is a YES instance if and only if (G, ϕ, ψ) is a YES instance. In one direction, it is clear that if (G', ϕ, ψ) is a YES instance, then so is (G, ϕ, ψ).

Now, suppose that (G, ϕ, ψ) is a YES instance. Then there is a path from ϕ to ψ in $\mathrm{Hom}(G, H)$ in which consecutive H-colourings differ on one vertex. It suffices to consider the case that ϕ and ψ differ on one vertex x, from which the general case follows. If $\phi\psi$ is an edge of $\mathrm{Hom}(G', H)$, then we are done; so, we assume that it is not. Since ϕ and ψ only differ on x, the only way that $\phi\psi$ can be an edge of $\mathrm{Hom}(G, H)$ and not $\mathrm{Hom}(G', H)$ is if x had no loop in G and $\phi(x) \nsim \psi(x)$. Observe that $\phi(x)\phi(y)\psi(x)$ is walk in H for any neighbour y of x in G'. If there are neighbours y and z of x such that $\phi(y) \neq \phi(z)$, then $\phi(x)\phi(y)\psi(x)\phi(z)$ forms a 4-cycle in H. Therefore, ϕ maps all neighbours of x to the same vertex of H. Let y be any neighbour of x in G (which exists, as x is not isolated). Let ϕ' be the map obtained from ϕ by changing the colour of x to $\phi(y)$. Then we have $\phi \sim \phi' \sim \psi$ in $\mathrm{Hom}(G', H)$. Thus, there is a recolouring between ϕ and ψ. □

2. Outline of proof

⟨sect:outline⟩ We assume, throughout the rest of the paper, that H is a triangle-free reflexive graph and that G is a reflexive graph. We also assume that G and H are connected; otherwise, we could simply deal with each component separately. We let ϕ and ψ be two H-colourings of G. Our goal is to present an algorithm to determine, in polynomial time, whether there is a recolouring between ϕ and ψ or, equivalently, whether there is a walk from ϕ to ψ in $\mathrm{Hom}(G, H)$.

Given a walk W from ϕ to ψ in $\mathrm{Hom}(G, H)$ and a vertex v of G, the sequence of vertices that v maps to under the homomorphisms of this walk is a walk from $\phi(v)$ to $\psi(v)$ in H. So, in order for a recolouring between ϕ and ψ to exist, it is necessary that there is a system $(W_v)_{v \in V(G)}$ of such walks W_v in H. Moreover, this system of walks must satisfy certain properties. For example, these walks are 'tethered' together in the sense that, if u and v are at distance d in G, then the ith vertices of W_u and W_v must be at distance at most d in H for all i. As it turns

out, the systems of walks which correspond to walks from ϕ to ψ can be completely characterized in terms of closed walks in G and their images under ϕ and ψ.

Given a closed walk C in G, a (ϕ, ψ)-walk W in $\mathrm{Hom}(G, H)$ induces a $(\phi(C), \psi(C))$-walk in $\mathrm{Hom}(C, H)$. This yields an immediate obstruction to the existence of a reconfiguration from ϕ to ψ. If there is no $(\phi(C), \psi(C))$-walk in $\mathrm{Hom}(C, H)$ for some closed walk C of G, then there can be no (ϕ, ψ)-walk in $\mathrm{Hom}(G, H)$. This obstruction manifests in two distinct ways.

The first is that $\phi(C)$ and $\psi(C)$ can be 'topologically' different in the sense that, even if we let them stretch (by adding extra vertices into C that map to the same target as their predecessor), we still cannot reconfigure $\phi(C)$ to $\psi(C)$. This happens, say, if $\phi(C)$ wraps around some big induced cycle of H and $\psi(C)$ does not. To formalize this, we use a graph $\Pi(H; r)$, (see Definition 3.1 and the pursuant discussion). This variation on $\mathrm{Hom}(C, H)$ is a discrete analogue of the fundamental group of a topological space; its vertices are all closed walks (of any length) of H starting at a fixed vertex r. We observe in Lemma 3.5 that a (ϕ, ψ)-walk W induces $(\phi(C), \beta_{W_r}(\psi(C)))$-walk in $\Pi(H; r)$ where r is the first vertex of $\phi(C)$, and $\beta_{W_r}(\psi(C))$ is a version of $\psi(C)$ that has been 'stretched out along the walk W_r.' So a necessary condition for the existence of a reconfiguration from ϕ to ψ will be that there is walk between versions of $\phi(C)$ and $\psi(C)$ in $\Pi(H; r)$. In Definition 4.1, a system (W_v) of walks is defined to be topologically valid if it describes such a walk for every closed walk of G.

The second way the obstruction manifests is based on the fact that a closed walk C of G can be 'tight' under ϕ. We define this in Definition 4.7, but it means that for any vertex v of C, ϕ maps the neighbours of v in C to different vertices of H. As H is triangle-free it is not hard to see that this implies that $\phi(C)$ has no neighbours in $\mathrm{Hom}(C, H)$; so v cannot move. Thus, in order for there to be a walk from ϕ to ψ, it is necessary that ϕ and ψ agree on all vertices in tight closed walks.

These two types of obstructions are analogous to two of the three obstructions identified by Wrochna [**?**] in the context of homomorphisms to C_4-free graphs. As we will show, in our setting, these are the only these two obstructions. That is, if all closed walks of G are topologically valid and ϕ and ψ agree on the tight closed walks, then there is a recolouring between ϕ and ψ. Therefore, proving Theorem 1.1 boils down to showing that we can either find a recolouring between ϕ and ψ or one of these two obstructions in polynomial time.

In Section 3 we define our discrete analogue $\Pi(H; r)$ of the fundamental group and prove, in strong analogy to basic homotopy theory, all the properties of it that we will need for our proof. In Section 4 we state a full technical version of Theorem 1.1 from which it clearly follows, and in Section 5 we prove this.

3. Necessary topological definitions and identities

⟨sect:top⟩ 3.1. **The Fundamental Graph.** For any integer $\ell \geq 1$ let P_ℓ be the reflexive path on $\ell + 1$ vertices. We refer to elements of $\mathrm{Hom}(P_\ell, H)$ as *walks* of H, and represent them as $(x_0 x_1 \ldots x_\ell)$. If $x_0 = a$ and $x_\ell = b$ the walk is an (a, b)-*walk*.

⟨def:moves⟩ **Definition 3.1.** Let $\Pi(H; a, b)$ be the graph whose vertices are (a, b)-walks in H, and in which a walk Y is adjacent to a walk $X = (x_0, \ldots, x_\ell)$ if either of the following are true:

(P1) $Y = (x_0, x_1, \ldots, x_{i-1}, x_i, x_i, x_{i+1}, \ldots, x_\ell)$ for any i, or

(P2) $Y = (x_0, x_1, \ldots, x_{i-1}, x'_i, x_{i+1}, \ldots, x_\ell)$ for some $i \notin \{0, \ell\}$, where $x_{i-1} = x_{i+1}$.

(Of course, Y is also adjacent to X if X is adjacent to Y via this definition.)

For an (a,b)-walk X,let $[X]$ be the component of $\Pi(H;a,b)$ containing X, and let $\pi(H;a,b)$ be the collection of these components.

For an (a,b)-walk X_1 and a (b,c)-walk X_2, let $X_1 \cdot X_2$ be the (a,c)-walk we get by identifying the first vertex of X_2 and the last vertex of X. Let $\overline{X_1}$ be the (b,a)-walk we get by reversing X_1. Clearly $\overline{X_1 \cdot X_2} = \overline{X_2} \cdot \overline{X_1}$. These are only partial operations on walks in H, but it is easy to see that if $Y_1 \in [X_1]$ and $Y_2 \in [X_2]$ then $\overline{Y_1} \in [\overline{X_1}]$ and $Y_1 \cdot Y_2 \in [X_1 \cdot X_2]$.

Write $\Pi(H;a)$ for $\Pi(H;a,a)$, and write $\pi(H;a)$ for $\pi(H;a,a)$. The operation '$\cdot$' is well defined on the set of closed walks with first vertex a, and one can show that it is well defined on $\pi(H;a)$. Thus, this operation on $\pi(H;a)$ is a group in which the inverse of $[X]$ is $[\overline{X}]$ and the identity is the component containing the length zero (a,a)-walk, which we denote by 0_a. We sometimes write this walk simply as 0 when the specific choice of basepoint is immaterial. The group $\pi(H;a)$ can be defined for graphs H that are not triangle free, though in this case the definition is more complicated. This is done is [**?**], where the authors go on to show that $\pi(H;a)$ is isomorphic to the fundamental group of the clique complex of H. Without formalising this, we simply view $\pi(H,a)$ as an analogue of the homotopy group of a space. In light of this analogy, we often refer to the first vertex a of a circuit as its *basepoint*, if $[C] = [C']$ we say that C and C' are *homotopic*, and if $[C] = 0$ we call it *contractible*.

We now observe some simple results about $\pi(H;a)$ that a reader familiar with homotopy theory will be familiar with. As we are defining them only for triangle-free graphs, their proofs are generally very easy.

It is easy to show, for (a,b)-walks X_1 and X_2, that

$$[X_1] = [X_2] \iff [X_1 \cdot \overline{X_2}] = 0. \tag{1}$$ eq:homwalks

Though the basepoint of a closed walk is fixed, we can change it. For an (a,b)-walk X and a b-closed walk C in $\pi(H;b)$ define the a-closed walk $\beta_X(C) := X \cdot C \cdot \overline{X}$. It is standard, and easily shown, that the map $\pi(H;b) \to \pi(H;a) : [C] \mapsto [\beta_X(C)]$ is a group isomorphism.

In particular, taking I as some initial segment of a closed walk C, it is not hard to see that the *cyclic shift* $\sigma_I(C)$ of C by $|I|$ vertices is in $[\beta_{\overline{I}}(C)]$, and so

$$[C] = [0] \iff [\sigma_I(C)] = [0]. \tag{2}$$ eq:perm

Thus $[C]$ is contractible if and only if all cyclic shifts of it are contractible. Breaking a cycle up as the composition of walks, this yield identities such as the following:

$$[X_1 \cdot X_2 \cdot X_3 \cdot X_4] = [0] \iff [X_2 \cdot X_3 \cdot X_4 \cdot X_1] = [0] \tag{3}$$?eq:perm2?

or using (1):

$$[\overline{X_1}] = [X_2 \cdot X_3 \cdot X_4] \iff [\overline{X_2}] = [X_3 \cdot X_4 \cdot X_1]. \tag{4}$$?eq:permedge?

There is one more simple observation which allows us to apply these notions succinctly.

Definition 3.2. When $C_1 = U_1 \cdot X \cdot V_1$ and $C_2 = U_2 \cdot \overline{X} \cdot V_2$ are closed walks as in Figure 1, let $C_1 \cdot_X C_2 = U_1 \cdot V_2 \cdot U_2 \cdot V_1$.

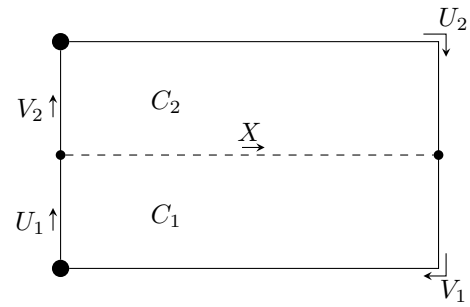

FIGURE 1. The cycle $C = C_1 \cdot_X C_2$ where $C_1 = U_1 \cdot X \cdot V_1$ and $C_2 = U_2 \cdot \overline{X} \cdot V_2$

⟨fig:sum⟩

With this definition, we have the following.

⟨lem:sum⟩ **Lemma 3.3.** *Let $C = C_1 \cdot_X C_2$. If two of $[C_1], [C_2]$ and $[C]$ are contractible, then they all are.*

Proof. Taking shifts $C_1' = V_1 \cdot U_1 \cdot X$ of C_1 and $C_2' = \overline{X} \cdot V_2 \cdot U_1$ we have that $[C_1' \cdot C_2'] = [C']$ where C' is the shift $V_1 \cdot U_1 \cdot V_2 \cdot U_1$ of $C = C_1 \cdot_X C_2$. It is easy to see therefore that two of $[C_1'], [C_2']$ and $[C']$ are contractible if and only if the third of them is. The result follows by (2). □

3.2. **Computation of reduced cycles.** For general H, it can be an NP-complete problem to decide whether $X' \in [X]$ for two closed walks X' and X. However, when H is triangle-free, it is easy. We say that a walk X in $\Pi(H; r)$ is *reduced* if it is a shortest walk in $[X]$.

⟨lem:redunique⟩ **Lemma 3.4.** *Any class $[X]$ in $\pi(H; r)$ has a unique reduced walk, and it can be found in linear time.*

Proof. We use the topological notion of a covering space. For a graph H and a vertex r, the *universal cover* $\mathcal{U}$ is the infinite graph whose vertices are walks in H starting at r, and having no occurrences of the same vertex at distance 1 or 2. Two walks X and Y are adjacent in $\mathcal{U}$ if we can get one from the other by dropping the last vertex. See Figure 2 for an example.

It is not hard to see that the map

$$\phi : \mathcal{U} \to H : (r, x_1, \ldots, x_\ell) \mapsto x_\ell$$

is locally bijective; i.e., induces a bijection between closed neighbourhoods $\{x\} \cup N(x)$ of vertices in $\mathcal{U}$ and vertices in H. It follows that the map

$$\Phi : (r, x_1, \ldots, x_\ell) \mapsto (\phi(r), \phi(r, x_1), \ldots, \phi(r, x_1, \ldots, x_\ell))$$

is a bijection between the set $\Pi(\mathcal{U}; (r))$ of walks in $\mathcal{U}$ starting at (r) and the set $\Pi(H; r)$ of walks in H starting at r. Now, two walks X, X' from (r) to $(r, x_1, \ldots, x_t)$ in $\mathcal{U}$ are adjacent in $\Pi(\mathcal{U}; (r), (r, x_1, \ldots, x_t))$ if and only if $\Phi(X)$ and $\Phi(X')$ are adjacent in $\Pi(H; r, x_t)$ and X is reduced if and only if $\Phi(X)$ is. So Φ induces a bijection on classes $[X]$ that is also a bijection on reduced walks. Since $\mathcal{U}$ is a tree, it is easy to see that a walk in $\Pi(\mathcal{U}; (r), (r, x_1, \ldots, x_t))$ is reduced if and only if it is the shortest walk between its endpoints in $\mathcal{U}$, so is unique in its class. Thus any class $[X]$ in $\pi(H; r)$ has a unique reduced walk.

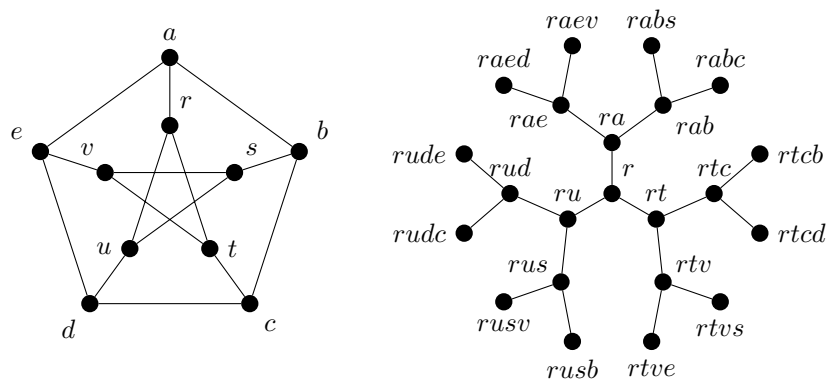

FIGURE 2. The Petersen graph and a finite portion of its universal cover.

⟨fig:UC⟩

The fact that the unique reduced walk in a class can be found in linear time is trivial, as if a walk is not reduced, we can always apply (P1) or (P2) to make it shorter.

□

3.3. **Homotopies of closed walks and the fundamental graph.** Recall that, if C is a closed walk with basepoint r, then a (ϕ, ψ)-path W in $\mathrm{Hom}(C, H)$ induces a $(\phi(r), \psi(r))$-walk W_r in $\mathrm{Hom}(r, H)$.

⟨lem:preswind⟩ **Lemma 3.5.** *If there is a (ϕ, ψ)-path W in $\mathrm{Hom}(C, H)$ for some cycle C with basepoint r, then $[\psi(C)] = [\beta_{W_r}(\phi(C)]$ in $\pi(H; \phi(r))$.*

Proof. Recall that, if there is a path from ϕ to ψ in $\mathrm{Hom}(C, H)$, then there is a path in which any two consecutive elements differ on one vertex. By induction on the length of this path, it suffices to consider the case that ϕ and ψ differ on only one vertex.

If they differ on any vertex but their basepoint, then there is nothing to prove, as then $[\psi(C)] = [\phi(C)]$ by (P2), and so we are done with empty W_r. So we may assume that they differ on the first vertex. That is, we have

$$\psi(C) = (x_0, x_1, \dots x_\ell, x_0) \text{ and } \phi(C) = (x_0', x_1, \dots x_\ell, x_0').$$

In this case take $W_r = (x_0, x_0')$, we will show that

$$[\psi(C)] = [(x_0, x_1, \dots, x_\ell, x_0)] = [(x_0, x_0', x_1, \dots, x_\ell, x_0', x_0)] = [\beta_{W_r}(\phi(C))].$$

Well, x_0, x_0' are adjacent (as C is reflexive) and are both adjacent to x_1 and $x_{\ell-1}$. So, as H is triangle-free, we have that $x_0' = x_1$ or $x_0 = x_1$. In the first case we have

$$[x_0, x_1, \dots, x_\ell, x_0] = [x_0, x_0' x_1, \dots, x_\ell, x_0]$$

by (P1) and in the second case we have it by (P2). Similarly $x_0' = x_\ell$ or $x_0 = x_\ell$, and we get the second equality that we need to finish the lemma by transitivity. □

If follows that there is a walk between ϕ and ψ in $\mathrm{Hom}(C, H)$ if and only if $[\phi(C)] = [\beta_W(\psi(C))]$ for some walk W between the basepoint of $\phi(C)$ and $\psi(C)$, but we would like to be able to find this W. The following allows us to do this.

3.4. **Basepoint-free homotopy.** Recall that adjacency in the graph $\Pi(H;r)$ is defined in Definition 3.1 by rules (P1) and (P2).

Definition 3.6. Let $\Pi(H)$ be the graph we get from the disjoint union, over all $r \in V(H)$, of $\Pi(H;r)$ by adding an edge between walks y and $x = (x_0, x_1, \ldots, x_{\ell-1}, x_0)$ if

(P3) $y = (x_1, \ldots, x_{\ell-1})$ and $x_1 = x_{\ell-1}$.

Let $\pi(H)$ be the set of components of $\Pi(H)$, and let $[\![C]\!]$ be the component of $\Pi(H)$ containing C. Clearly we have that

$$[C] = [C'] \text{ implies } [\![C]\!] = [\![C']\!]. \tag{5}$$ eq:equivhom

The converse is not true in general, indeed not even heuristically, as C and C' may satisfy $[\![C]\!] = [\![C']\!]$ and have different basepoints. However, if we have a path W from C to C' in $\Pi(H)$, then it is not hard to see that $[C] = [\beta_{W_r}(C')]$ where W_r is the trace of the basepoint r of C,

If C is a cycle with basepoint r, then it is trivial to see that $[\![C]\!] = [\![\beta_P(C)]\!]$ for any walk P ending at r, and so from Lemma 3.5 we get that if there is a (ϕ, ψ)-path in $\mathrm{Hom}(C, H)$ then $[\![\phi(C)]\!] = [\![\psi(C)]\!]$. A closed walk C is *free-reduced* if it is a shortest closed walk in $[\![C]\!]$. Though any component in $\pi(H;r)$ contains a unique reduced closed walk, this does not hold for free-reduced closed walks in components of $\pi(H)$. Indeed, if C is free-reduced, any cyclic shift of it is also free reduced. However, any non-contractible reduced closed walk C, 'contains' a unique free-reduced closed walk.

⟨fact:redcycle⟩ **Fact 3.7.** *If C is a non-contractible reduced cycle with basepoint r, then C decomposes uniquely as $C = \beta_T(C_f)$ for some free reduced cycle C_f, and some reduced walk T from r to the first vertex of C_f.*

Proof. Indeed, as $C = (x_0, x_1, \ldots, x_{\ell-1}, x_0)$ is reduced with respect to operations (P1) and (P2) the only possible reduction is operation (P3), meaning $x_{\ell-1} = x_1$ and it reduces to $C_1 = (x_1, x_2 \ldots, x_{\ell-1})$. Where $T_1 = (x_0, x_1)$ we have $C_1 = \beta_{T_1}(C_1)$. By induction we get the fact. □

As the reduced form of C is unique and can be found in linear time, so can its decomposition into $\beta_T(C_f)$. We call this its *free decomposition* and call T its *tail*.

This free decomposition does not change by much under basepoint change. To show this we start with a somewhat technical calculation. Intuition for this lemma is perhaps aided by first reading the corollaries that follow it.

A walk I is *initial* in a walk X if $X = I \cdot J$ for some J. A walk is *terminal* in X if it is initial in $\overline{X}$.

Lemma 3.8. *Let A and B be be a non-contractible reduced cycles with respective free decompositions $\beta_T(A_f)$ and $\beta_S(B_f)$. If $[A] = [\beta_W(B)]$ for a walk W then there is an integer d and initial I of A_f^d such that $B_f = \sigma_I(A_f)$ is the cyclic shift of A_f, and $[W] = [T \cdot I \cdot \overline{S}]$. This is reduced unless T and S end in a common walk.*

Proof. As $[A] = [\beta_W(B)]$ we have $[A^d] = [\beta_W(B^d)]$ so $[A^d W] = [W B^d]$ for any integer d. While $A^d W$ and $W B^d$ are not generally reduced, A^d and B^d reduce to $\beta_T(A_f^d)$ and $\beta_S(B_f^d)$, and so by taking d large enough, we can say $A^d W = W B^d$ reduces to

$$I_A \cdot T_W = I_W \cdot T_B \tag{*}$$ eq:com

for some initial I_A of $\beta_T(A_f^d)$, some termimal T_W of W, some initial I_W of W and some terminal T_B of $\beta_S(B_f^d)$. As (*) holds for all large enough d, and I_A and T_B grow with d, we have that $|A_f| = |B_f|$, and that by taking d large enough, I_A and T_B can be assured to have an arbitrarily long overlap, so B_f is some shift of A_f, as needed. We now just have to verify the decomposition of W.

Again from (*) for large enough d we get that the end T_W of W is terminal in T_B so in $B_f^d \cdot \overline{S}$, and as $\beta_T(A_f^d) \cdot W$ reduced to $I_A \cdot T_W$, we get that $W = W' \cdot T_W$ for some W' which cancels with the end of $A_f^d \cdot \overline{T}$. Thus W' is initial in $T \cdot A_f^{-d}$.

Now, W' ends where T_W starts. If T and S end in a common walk, then it is possible that $W' \cdot T_W$ is the reduction of $T \cdot S$. Otherwise W' contains T and T_W contains S. In this case W' begins with T and then contains some initial walk of A_f^{-d}. So $W' \cdot T_W$ is $T \cdot I \cdot \overline{S}$ for some initial I of A_f^d or A_f^{-d}, as needed. That $B_f = \beta_I(A_f)$ now follows from the fact that B_f is a shift of A, and that I is an initial walk in A_f or its inverse that has the right endpoints. □

From this we get a couple of immediate corollaries.

⟨cor:freeredcycle⟩ **Corollary 3.9.** *If A_f and B_f are free reduced non-contractible cycles with $[\![A_f]\!] = [\![B_f]\!]$ then B_f is a cyclic shift of A_f.*

Proof. As $[\![A_f]\!] = [\![B_f]\!]$ we have $B_f \in [\beta_{W_r}(A_f)]$ where W_r is the trace of the basepoint r under the walk W from A_f to B_f in $\Pi(H)$. By the lemma B_f is a shift of A_f, □

For a reduced cycle C let $\sqrt{C}$ be the shortest initial walk of C such that $\sqrt{C}^d = C$ for some positive d.

⟨cor:formhomotopy⟩ **Corollary 3.10.** *If A and B are non-contractible reduced cycles free decompositions $A = \beta_T(A_f)$ and $B = \beta_S(B_f)$, then $[A] = [\beta_W(B)]$ if and only if W is of the form*

$$W_d = \sqrt{A}^d \cdot T \cdot I \cdot \overline{S}$$

for some integer d, where I is the initial walk of A from the end of T to the beginning of $\overline{S}$.

Proof. The 'if' part is trivial as $[W_d] = T \cdot \sqrt{A_f}^d \cdot I \cdot \overline{S}$ and this is the trace of the path from A to B that we get by composing the reduction of A to A_f with the cyclic shift of A_f by $A_f^d \cdot I$ to B_f, and then composing this with the inverse of the reduction of B to this shift of B_f. The 'only if' is from the lemma. □

⟨cor:uniqpath⟩ **Corollary 3.11.** *Let A and B be reduced cycles in G with basepoint r having respective free decompositions $\beta_T(A_f)$ and $\beta_S(B_f)$, and such that $\sqrt{A} \neq \pm\sqrt{B}$. If there exists a walk W_r such that $[\phi(A)] = [\beta_{W_r}\psi(A)]$ and $[\phi(B)] = [\beta_{W_r}\psi(B)]$, then there is a unique reduced such walk and it is the reduction of either $T \cdot I \cdot \overline{S}$ for some initial I of A_f or $T' \cdot I' \cdot \overline{S'}$ for some initial I' of B_f.*

Proof. Let A and B and W be as in the premise of the corollary. By Corollary 3.10 we have then

$$[\sqrt{A}^d \cdot T \cdot I \cdot \overline{S}] = [\sqrt{B}^{d'} \cdot T' \cdot I' \cdot \overline{S'}]$$

for some d and d' where I and I' (and all other components) are fixed. Reducing these, we get

$$[T \cdot \sqrt{A_f}^d \cdot I \cdot \overline{S}] = [T' \cdot \sqrt{B_f}^{d'} \cdot I' \cdot \overline{S'}]$$

where A_f is the free reduction of A and B_f is the free reduction of B. As $\sqrt{A} \neq \sqrt{B}$ we have either $T \neq T'$ or $\sqrt{A_f} \neq \sqrt{B}_f$. Either way, the only possible solution has $d = 0$ or $d' = 0$. These cannot yield different solutions by considerations of length. □

4. Topologically valid systems of walks and tight cycles

⟨sect:TV⟩ In this section we finally define our obstructions to the existence of a reconfiguration from ϕ to ψ.

⟨def:TV⟩ **Definition 4.1.** A *system of walks for ϕ and ψ* is a vector $(W_v) = (W_v)_{v \in V(G)}$ of walks in H such that W_v is $(\phi(v), \psi(v))$-walk. It is *topologically valid* if for every closed walk C of G we have

$$[\phi(C)] = [\beta_{W_r}(\psi(C))] \tag{6}$$?eq:TV?

Observe that this definition is not affected by assuming that all walks in the system (W_v) are reduced, so we will often do this. To determine if a system of walks for ϕ and ψ is topologically valid, it will be useful to observe an alternate defintion.

Generalizing the condition for closed walks, a uv-walk U in G is *(topologically) preserved* by a system (W_v) of walks for ϕ and ψ if

$$[\phi(U)] = [W_u \cdot \psi(U) \cdot \overline{W_v}]. \tag{7}$$ eq:TV1

By (1) and (2) we can write this in either of the following forms:

$$[\phi(U)] = [W_u \cdot \psi(U) \cdot \overline{W_v}] \tag{7'}$$ eq:TV2

$$[\phi(U) \cdot W_v \cdot \overline{\psi(U)} \cdot \overline{W_u}] = 0 \tag{7''}$$ eq:TV3

Now, for a uw-walk U and a wv-walk V we have

$$\phi(U \cdot V) \cdot W_v \cdot \overline{\psi(U \cdot V)} \cdot \overline{W_u} = \left(\phi(U) \cdot W_w \cdot \overline{\psi(U)} \cdot \overline{W_u}\right) +_{W_w} \left(\phi(V) \cdot W_v \cdot \overline{\psi(V)} \cdot \overline{W_w}\right),$$

so by Lemma 3.3 and form (7″), we get that if two of U, V and $U \cdot V$ are preserved by (W_v), then so is the third of them. This yields the following alternate definitions.

?⟨fact:TV'⟩? **Fact 4.2.** *A system of walks (W_v) is topologically valid for ϕ and ψ if and only if every edge in G is topologically preserved by (W_v) if and only if every walk in G is toplogically preserved by (W_v).*

⟨fact:necTV⟩ **Fact 4.3.** *Let W be a reconfiguration from ϕ to ψ in* $\mathrm{Hom}(G, H)$*. Where for every vertex v of G, W_v is the trace of v, the system (W_v) of walks is topologically valid for ϕ and ψ.*

Proof. In Section 2 we observed that W induces a $(\phi(C), \psi(C))$-walk in $\mathrm{Hom}(C, H)$ for any closed walk C of G. In Lemma 3.5 we observed that where r is the basepoint of C this implies $[\psi(C)] = [\beta_{W_r}(\phi(C)]$ in $\pi(H; \phi(r))$. Thus any reconfiguration W from ϕ to ψ yields a system (W_v) of walks that is topologically valid for ϕ and ψ. □

The topologically valid system of walks (W_v) we get from W is not generally reduced, each walk in general have many repeated vertices.

?⟨def:realizable⟩? **Definition 4.4.** A system of walks (W_v) is *realizable* if by adding repeated vertices into walks we get a system of walks that is induced by a reconfiguration.

?⟨lem:uniquesystem⟩? **Lemma 4.5.** *Let W_r be a fixed walk in H from $\phi(r)$ to $\psi(r)$. If there is a topologically valid system (W_v) of reduced walks for ϕ and ψ in* $\mathrm{Hom}(G, H)$*, then it is unique.*

Proof. Let W_r be a reduced $(\phi(r), \psi(r))$-walk in H, and let (W_v) be a topologically valid system of walks containing W_r. Observe that for a neighbour r' of r, the requirement that the edge $e = r'r$ is preserved under (W_v) means, using form $(7')$, that $W_{r'}$ is the reduction of $\phi(r'r) \cdot W_r \cdot \overline{\psi(r'r)}$. By the connectedness of G it follows that (W_v), if it exists, is unique. □

In light of this fact, we make the following definition

Definition 4.6. A $(\phi(r), \psi(r))$-walk W_r is *topologically valid* for ϕ and ψ if it is contained in a system $(W_v) = (W_v)_{v \in V(G)}$ of walks that is topologically valid for ϕ and ψ. If W_r is topologically valid, it *generates* the system (W_v).

⟨def:tight⟩ **Definition 4.7.** A cycle C of G is *tight* under ϕ if $\phi(C)$ is a free reduced in $\pi(H)$. Any vertex of a tight cycle is *fixed* under ϕ.

It is not hard to see that if C is tight under ϕ then any neighbour of ϕ in $\mathrm{Hom}(G, H)$ must agree with ϕ on C. Thus a fixed vertex is one that cannot change under a reconfiguration. A walk is *constant* if all vertices are the same.

⟨fact:nectight⟩ **Fact 4.8.** *If the system of walks (W_v) is realizable for ϕ and ψ in* $\mathrm{Hom}(G, H)$ *and c is fixed under ϕ, then W_c is constant.*

5. Full statement and proof of the main theorem

⟨sect:proof⟩ In this section we prove the following theorem, from which Theorem 1.1 follows immediately.

⟨thm:mainT⟩ **Theorem 5.1.** *Let H be a reflexive symmetric triangle-free graph, and (G, ϕ, ψ) be an instance of* $\mathrm{Recol}(H)$*. There is a reconfiguration from ϕ to ψ if and only if there is a system of walks $(W_v)_{v \in V(G)}$ that is topologically valid for ϕ and ψ, such that W_c is a constant walk for any vertex c in cycle C that is tight under ϕ.*

Moreover, these validity of this conditions can be determined by an algorithm that runs in time that is polynomial in $|V(G)|$.

The 'necessity' part of this theorem already done in Facts 4.3 and 4.8. The 'suffciency' part is not much harder; it will follow from Algorithm 2. The harder part is deciding if there is such a system of walk. For this we present a series of algorithms. We formally put the proofs all parts together at the end of the section.

For the following algorithms, H is a fixed triangle-free graph. All Algorithms are about a given (G, ϕ, ψ) of $\mathrm{Recol}(H)$, and running times are given in terms of $|V(G)|$. The vertex r is a fixed vertex of G that we implicitly take as the basepoint of cycles.

⟨alg:TV⟩ **Algorithm 1.** *Let W_r be reduced walk from $\phi(r)$ to $\psi(r)$. We can determine in polynomial time if W_r is topologically valid. If it is, we produce the system (W_v) it generates, if not, we provide a closed walk B of G such that $[\phi(B)] \neq [\beta_{W_r}(\psi(B)]$.*

Proof. To determine if W_r is topologically valid, choose any spanning tree T of G with root r. Inductively build (W_v) by setting W_v to be the reduction of $\phi(vu) \cdot W_u \cdot \overline{\psi(vu)}$, where the $(\phi(u), \psi(u))$-walk W_u is already defined for the precursor u of v in T.

By construction any edge in T, so any path in T is preserved by (W_v). Thus the system (W_v) is topologically valid if and only if every edge e in $G \setminus T$ is also preserved by (W_v). This can be determined in polynomial time.

Now, in the case that (W_v) is not topologically valid, we found some edge $e = uv$ that is not preserved by (W_v). Letting T_v denote the unique path in T from r to v, we clearly have then that the closed walk $B = T_u \cdot e \cdot \overline{T_v}$ is not preserved by (W_v), so $[\phi(B)] \neq [\beta_{W_r}(\psi(B))]$. □

⟨alg:realizable⟩ **Algorithm 2.** *For a given system (W_v) of paths that is topologically valid for given $\phi, \psi \in \mathrm{Hom}(G, H)$, we can decide in time linear in the sum of the lengths of the paths if (W_v) is realizable. If it is not realizable, we give a closed walk C on which ϕ is tight, and on which a) ϕ and ψ disagree, or b) for every vertex v of C, W_v is a non-empty closed walk.*

Proof. Let homomorphisms $\phi, \psi : G \to H$ and a system of walks $(W_v)_{v \in G}$ that is topologically valid for ϕ and ψ be given. Starting with ϕ we attempt to construct a reconfiguration W from ϕ to ψ, by changing one vertex v at a time from its current position W_v to its next position. If we succeed, this witnesses the realizability of (W_v). As any edge $e = uv$ of G is topologically valid under the system of walks,

$$[W_u] = [\phi(e) \cdot W_v \cdot \overline{\phi(e)}],$$

and as by Lemma 3.4 there is a unique reduced walk in $[W_u]$, we have, where a superscript of i on the walk designates that i^{th} vertex of the walk, either $W_u^i = W_v^{i-1}$ for all i except maybe $i = |V(W_u)|$, or $W_v^i = W_u^{i-1}$ for all i except maybe $i = |V(W_v)|$.

Construct an auxillary digraph A on $V(G)$ by setting $u \to v$ if

$$W_v^1 = W_u^0 \text{ and } W_v^0 \nsim W_u^1. \tag{8}$$ eq:aux

(Interpret this as 'u wants to move to W_u^1, but v has to move first'.) Observe that if uv is an edge, and $|W_u| = 0$, then $|W_v| \leq 1$, and $W_v^1 \sim W_u^0$, so u has no arcs in A.

If A has any sink v, then on moving v to W_v^1, we still have a homomorphism. Append this homomorphism to W, and relabelling W_v by removing its first vertex. As each relabelling reduces the length of a walk in (W_v) by one, we either complete in time linear in the sum of the lengths of the paths, (and we are done) or at some step A has no sinks, so contains a directed cycle.

Assume that A has a directed cycle $C = a_1 \to a_2 \to \cdots \to a_t$. By construction, every arc $a_i \to a_{i+1}$ of C is an edge $a_i \sim a_{i+1}$ of G, and (8) ensures that $\phi(a_i) \notin \{\phi(a_{i-1}), \phi(a_{i-2})\}$, (indices modulo t) for all i. Thus is C is a tight cycle under ϕ.

If ϕ and ψ do not agree on C, then we are done. If they agree then for any vertex v of C, then W_v is a closed walk, and non-empty or v would have no arcs in A. □

We observe now one of the main steps of our overall decision algorithm.

⟨alg:Step2⟩ **Algorithm 3.** *Give a walk $(\phi(r), \psi(r))$-walk W_r that is topologically valid for ϕ and ψ in $\mathrm{Hom}(G, H)$ we can decide polynomial time (in $|V(G)| + |W_r|$) if there is any realizable topoligically valid $(\phi(r), \psi(r))$-walk.*

Proof. Given W_r, apply Algorithm 1, in polynomial time, to generate from W_r the unique topologically valid system (W_v) of walks that contains it. The apply Algorithm 2 to decide, in polynomial time, if it is realizable. It it is, we are done. So we may assume that it is not, and Algorithm 2 returns us a closed walk C that is tight under ϕ, and on which ϕ and ψ agree. As any vertex u of C is fixed under ϕ the only possible topologically valid $(\phi(u), \psi(u))$-walk W_u is the empty walk on $\phi(u)$. Apply Algorithm 1 to decide if this is topologically valid, if it is not, then there are no topologically valid systems of paths, and we are done. So we may assume that it is topologically valid. Apply Algorithm 2 to decide if it is realizable. Whatever the outcome, we are done. □

What is left to do, is to decide if there is a topologically valid path W_r. We need one more observation to help us do this.

⟨alg:Step1⟩ **Algorithm 4.** *We can determine in polynomial time if there is a $(\phi(r), \psi(r))$-walk that is topologically valid for ϕ, ψ in* $\mathrm{Hom}(G, H)$*. It there is, it is provided.*

Proof. Let W_r be any shortest $(\phi(r), \psi(r))$-walk in H, and apply Algorithm 1 to determine if it is topologically valid. If it is, we are done, so we may assume that it is not, and the algorithm returns a closed walk C of G such that $[\phi(C)] \neq [\beta_{W_r}(\psi(C))]$, so necessarily $\phi(C)$ is non-contractible. We use this C to help us decide if there is any other $(\phi(r), \psi(r))$-walk W that is topologically valid.

By Lemma 3.4, (and Fact 3.7) we can find the free decompositions $\phi(C) = \beta_T(C_f)$ and $\psi(C) = \beta_S(C'_f)$ in polynomial time. We can check in polynomial time if C'_f is a shift of C_f. If it is not, then $[\![\phi(C)]\!] \neq [\![\psi(C)]\!]$ by Corollary 3.9, and so by (5), we have

$$[\phi(C)] \neq [\beta_W(\psi(C))]$$

for any W. Thus we answer NO and are done.

We may assume that C'_f is the shift $\sigma_I(C_f)$ for some initial walk I of C_f. By Corollary 3.10 we thus have that the only reduced walks W that preserve C are reductions of

$$W_d := \sqrt{\phi(C)}^d \cdot T \cdot I \cdot \overline{S}$$

for integers d.

Apply Algorithm 1, to decide if W_0 is topologically valid. If it is we are done, so we may assume that it is not, and the algorithm returns another closed walk C' with $[\phi(C')] \neq [\beta_{W_0}(\psi(C'))]$. By Corollary 3.10 we get that $\sqrt{C'} \neq \pm\sqrt{C}$ and so by Corollary 3.11 the only walks W that can preserve both C and C' are the reductions of $T \cdot I \cdot \overline{S}$ and $T' \cdot I' \cdot \overline{S'}$ where $\beta_{T'}(C_f)$ is the reduction of $\phi(C')$ and $\beta_{S'}(\sigma_{I'}(C_f))$ is the reduction of $\psi(C')$.

Both of these have length at most $2n$ so we can check them for topological validity in polynomial time using Algorithm 1. □

Though it should be clear, we now give the formal proof of Theorem 5.1.

Proof of Theorem 5.1. Let an instance (G, ϕ, ψ) of $\mathrm{Recol}(H)$ for triangle-free H be given. As mentioned above, 'necessity' is done in Facts 4.3 and 4.8.

For 'sufficiency', assume we have a system of walks (W_v) that is topologically valid for ϕ and ψ that is constant on all vertices in a tight cycle. Applying Algorithm 2 we must get a realization of (W_v) as otherwise the Algorithm provides a tight

cycle C on which the reduction of W_c for any of its cycles in non-empty, so W_c is non constant.

To decide if there is a such a realizable system of walks, we apply Algorithm 4 to determine if there is a topologically valid system of walks. If there is not, then we are done, if there is, we apply Algorithm 3 to determine if there is one that is realizable. □

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